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MILITARY SPECIFICATION

ELECTRON TUBE, CW FIELD TUNED OSCILLATOR

JAN-7533

The complete requirements for procuring the electron tube described herein shall consist of this document and the issue in effect of MIL-E-1.

DESCRIPTION: UHF, Pencil Triode — Integral Cavity CW Oscillator F = 1660–1700 Mc

DIMENSIONS: See figure 1.

ABSOLUTE RATINGS:

Parameters:	E_f	E_b	I_b	P_p	t_k	TA	Alt.
Unit:	V	Vdc	mAdc	W	sec	°C	ft.
Maximum:	6.6	117	34	4.0	.....	+75	100,000
Minimum:	.....	.....	.....	.....	60	—55	.....

TEST CONDITIONS:

Parameters:	E_f	E_b	t_k	R_g
Unit:	V	Vdc	sec	ohms
				2400 max
	5.2	95	180	1300 min

FSC 5960

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Paragraph No.	Test	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
						Min	Max	
	General							
3.1	Qualification	Required	.....	.....	.....	.....	.....	.....
3.6	Performance	(See note 1)	.....	.....	.....	.....	.....	.....
4.5	Holding period	.....	.....	.....	.....	96	.....	Hrs
4.9.2	Dimensions	(See fig. 1)	.....	.....	.....	.....	.....	.....
	Acceptance inspection part 1 (production) (see notes 2 and 3)							
4.10.8	Heater current	Ef = 5.2V	0.65	II	If	135	157	mA
.....	Frequency	Ef=5.2V Eb=95Vdc; circuit per dwg. 169 JAN; Z_L =50 ohms; VSWR=1.1:1 max. (see note 4).	0.65	II	F	1676	1684	Mc
4.10.2.2	Power oscillation (1).	Same as frequency test (see notes 5, 6, and 7).	0.65	II	Po(1)	270	.....	mw
4.10.2.2	Oscillator plate current (1).	Ef=6.6V; Eb=117 Vdc; F=1680 Mc; circuit per dwg. 169 JAN; Z_L=50 ohms; VSWR=1.1:1 max; (see notes 5, 6, and 7).	0.65	II	Ib(1)	.....	34	mAdc
.....	Plunger pressure	No voltages (see note 8).	0.65	II	F Po	 270	±4	Mc mW
	Acceptance inspection part 2 (design)							
4.7.3.1	Tap shorts	(See note 9)	2.5	II	.....	.....	.....	.....
4.10.2.2	Oscillator plate current (2).	Same as Osc. Ib(1) except F=1660 $+\frac{3}{-1}$ Mc (see note 7)	2.5	II	Ib(2)	.....	34	mAdc
4.10.2.2	Power oscillation (2).	Same as pwr. osc. (1) except F=1660 $+\frac{3}{-1}$ Mc (see note 7).	2.5	II	Po(2)	250	.....	mW
4.10.2.2	Oscillator plate current (3)	Same as osc. Ib(1) except F=1700 $+\frac{3}{-1}$ Mc (see note 7).	2.5	II	Ib(3)	.....	34	mAdc
4.10.2.2	Power oscillation (3)	Same as pwr osc. (1) except F=1700 $+\frac{3}{-1}$ Mc (see note 7).	2.5	II	Po(3)	250	.....	mW
4.10.7.3.2	Tuning range	Power oscillation (1)	6.5	E	ΔF	±20	.....	Mc
4.8	Electrode insulation	Ef=6.3; E=100 (grid neg.) Ef=6.3; E=300 (grid neg.).	6.5 6.5	E E	Rgk Rgp	25 25	....	meg meg
.....	Vibration	No voltages (see note 10).	6.5	E	.....	.....	.....	.....
	Dimensions	(See fig. 1)	.....	.....	.....	.....	.....	.....

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Paragraph No.	Test	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
						Min	Max	
	Acceptance inspection part 3 (periodic or life tests)							
	Frequency drift and jump.	F=1680±4 Mc (see notes 11 and 12).	.....	.....	F	.....	+4 −1	Mc
	Power oscillation (4) (mismatched load).	Eb=95 Vdc; Ef=5.2V; Rg=560 ohms, min, 2400 ohms max; Z_L=50 ohms; VSWR 1.05; F=1680 Mc; tk=15 min, minimum (see notes 11 and 13).						
		Pulling factor; VSWR =1.2.1.	.....	.....	ΔF	.....	6.0	Mc
		Power output: VSWR =1.2:1.	.....	.....	Po	210	.....	mW
4.9.12.1	Low-pressure voltage breakdown.	Pressure=8±3 mmHg E=200 Vac (see notes 11 and 14).	.....	.....	.....	.....	.....	.....
	Dimensions	(See fig. 1)	.....	.....	.....	.....	.....	.....
4.9.5.3	Lead fatigue	(See notes 11 and 15)	.....	.....	.....	2	.....	arcs
	Tuning torque	(See notes 11 and 16)	.....	.....	.....	0.25	6.5	In-Oz

Paragraph No.	Test	Conditions	Allowable defects per characteristic		Symbol	Limits		Units
			1st sample	Combined sample		Min	Max	
4.11	Life test	Ef=6.6; Eb=117Vdc; Rg/Ib=34 mAdc; F=1680 ± 4 Mc (see note 17).	.....	.....	.....	.....	.....	.....
4.11.4	Life test end points (15 minutes).	Power oscillation (1)	1	2	Po(1)	270	.....	mW
		Change in pwr. osc. (1) (individual units) (see note 18).			$\frac{-\Delta Po(1)}{t}$	.....	30	%
4.11.4	Life test end points (5 hours).	Power oscillation (1)	2	6	Po(1)	210	.....	mW
		Change in pwr. osc. (1) (individual units) (see note 18).			$\frac{-\Delta Po(1)}{t}$	.....	30	%
		Total defectives	4	12	.....	.....	.....	.....
4.9.18	Container drop.	(d) Package group 1; Carton size H Power oscillation (1)	.....	.....	ΔF	.....	±4	Mc

NOTES

1. In addition to the paragraphs specified hereon, the following tests and requirements in 3.6 apply: 3.3, 3.3.1, 3.4.1, 3.7, 3.7.7, 3.8, 4.1, 4.3, 4.4, 4.5, 4.6, 4.8, 4.9.1, 4.9.2, and 4.9.21.

2. The AQL for the combined defectives for attributes in acceptance inspection, part 1, shall be 1%. A tube having one or more defects shall be counted as one defective. Standard MIL-STD-105, inspection level II, shall apply. If any individual characteristic exceeds 0.65% AQL, and also the entire group fails the

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combined 1.0% AQL, the lot need only be detailed 100% for the individual failing characteristic, prior to resubmission. However, if the lot failure results from defects of two or more characteristics, detailing shall include all characteristics in the combined AQL group.

3. Use single, double or multiple sampling provisions per MIL-STD-105.

4. At the time of shipment all cavity units will be tuned to 1680 ± 4 Mc.

5. After a 96-hour holding period and without further adjustments, the frequency and power output shall be within the limits specified. If 100% retesting is required for the power oscillation (1) test, the 96-hour hold period shall be repeated.

6. Power oscillation (1) shall not be below the specified limit and oscillator plate current (1) shall not exceed the specified limit when a grid resistor within the specified range is used.

7. The choice of grid resistor should be such that for any individual tube, the oscillator plate current should not exceed the specified maximum limits and the power oscillation should not fall below the specified minimum limits over the frequency range from 1660 to 1700 mc.

8. Subject tubes to a pressure of 6 pounds minimum applied to the cathode and anode plungers. There shall be no perceptible movement of the plungers during the test and the tubes shall meet the power output and frequency limits after pressure test.

9. Tubes shall be tested per MIL-E-1D, paragraph 4.7.3.1 except the indicating device shall have a resistance vs. time duration curve meeting the following points.

600,000 ohms — permanent short
500,000 ohms — 500 microseconds
100,000 ohms — 100 microseconds
1,000 ohms — 60 microseconds

The applied voltage between adjacent elements of the tube under test shall have a value between 20 and 70 volts dc or peak ac.

The tube shall be placed in an adapter and mounted on a paddle. The paddle shall be tapped with a rubber tapper three times in two planes 90° to 120° apart. If a short indication is obtained, the tapping cycle shall be repeated two more times. If a short indication is again obtained, the tubes shall be rejected as inoperable.

10. The cavities shall be vibrated in two planes; perpendicular and parallel to the major axis $F = 50$ to 60 cps; $t = 3$ minutes; $g = 10$; no voltages. At the conclusion of the vibration test, the cavities must meet the specified limits for tuning range.

11. This test shall be performed as a periodic check test. Tubes for periodic check tests shall be selected at random at intervals so that when the tube is in production, at the time any lot is offered for acceptance, not less than 6 tubes will have been checked in the last six calendar months. Should the tubes fail such a test, that test will become a design test (6.5% AQL, Insp. Level L6) until three consecutive lots pass. Thereafter the test will again become a periodic test.

12. Operate for 5 minutes with $E_f = 6.6$ V; $E_b = 117$ Vdc; $R_g/I_b = 34$ mAdc max.; $T = +22°$ C. (approx.), after which the frequency shall be recorded. The ambient temperature shall then be reduced to =40° C. during a 30-minute operating period. Simultaneously and gradually, both heater and plate voltages shall be reduced so that the last 15 minutes interval shall be at $E_f = 5.2$ V, $E_b = 95$ Vdc. The change in frequency from the initial to final value shall be within the limits specified. There shall be no change in frequency of more than 2 mc in any 15 seconds throughout the test.

13. Test shall be performed in accordance with Dwg. 250 JAN with the following exceptions:

Initially adjust Rg to meet the conditions of the oscillator plate current (1) test. Without adjustment of Rg, decrease voltages to the specified values; the power input to the tube attained under these conditions shall be kept constant throughout the test.

(a) The load shall be varied through all phase angles at a VSWR of 1.2:1 only.

(b) The data obtained will not be plotted on a Smith Chart.

14. The specified voltage shall be applied between the anode and the grid plus cathode. No other

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voltages shall be applied. Tubes showing evidence of corona or arcing shall be considered defective. This is a nondestructive test.

15. The free heater lead only shall be subjected to this test. Subject the lead to a 16 ± 1 oz. pull. The lead must withstand the specified number of 90° arcs at the tube base. (An arc is described as the movement of the lead away from the tube and back to normal). A lead fails at the time it breaks off from the tube. This is a destructive test. Electrical rejects may be used.

16. The tightness of the frequency tuning screws shall be measured with a standard torque screw driver. The torque required to tune the cavity from 1660 to 1700 Mc. shall be within the limits specified.

17. Adjust Rg for plate current of 34 mA initially only. Subsequent tests on the units (at 15 min. and at 5 hrs.) shall be made using the initial value of Rg.

18. The life test sample shall consist of 40 tubes per week. In the event of failure of the first sample on life test, the second sample shall consist of 80 tubes. Acceptance shall then be based on results of the combined first and second samples. This is a nondestructive test. Cavities which pass intial test limits of acceptance inspection, part 1 (production) shall be accepted under this specification.

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Army—EL
Navy—Ships
Air Force—ASD

Preparing activity:
Army—EL

Project No. 5960-1637

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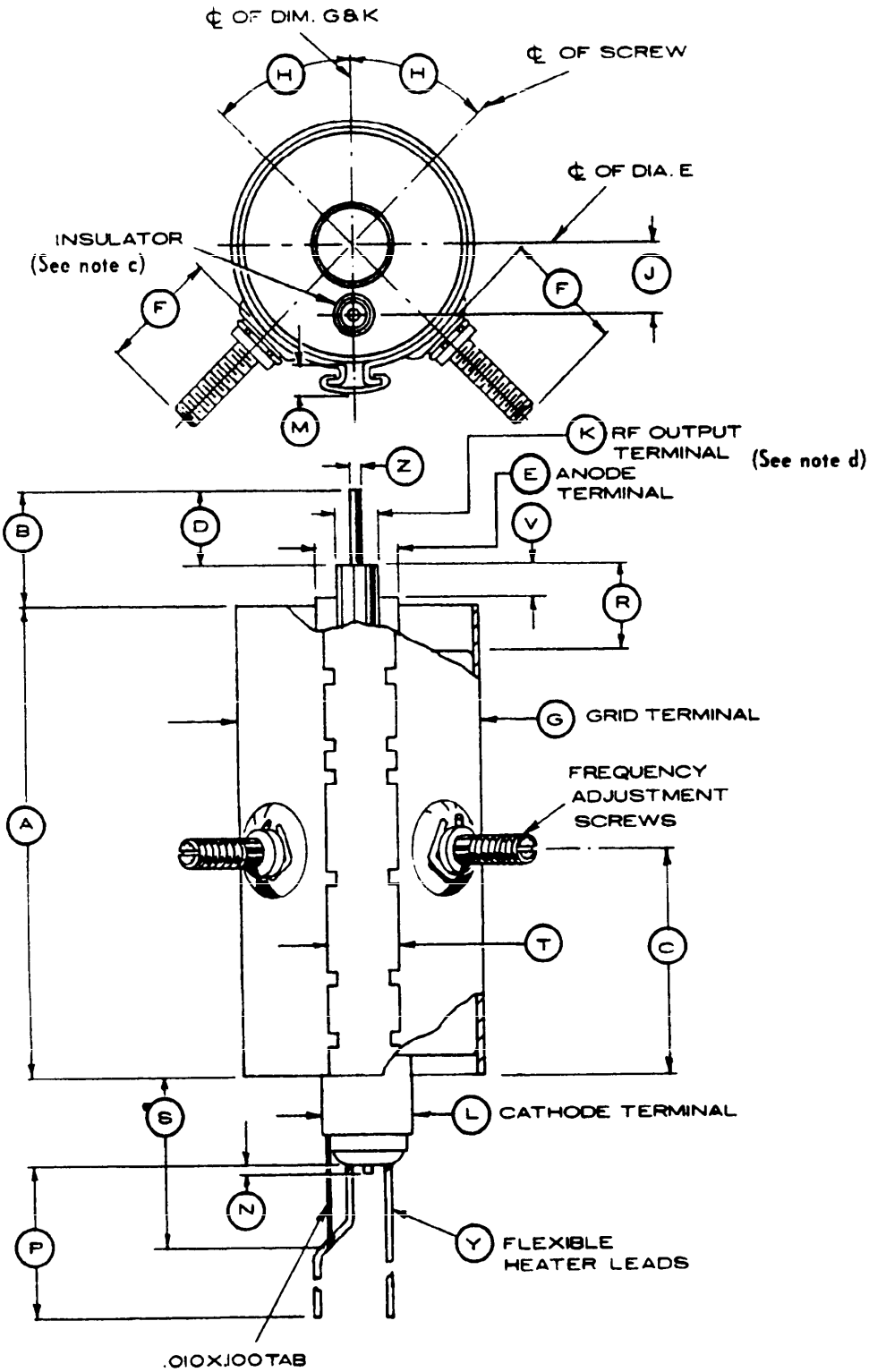


Figure 1. Outline Drawing

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Dim.	AQL %	Inspection level	Limits			
			Inches		Millimeters	
			Min	Max	Min	Max
A	Note b	Code G	.....	1.730	.....	43.94
B			.....	0.600	.....	15.24
C			0.777	0.817	19.74	20.75
D			0.221	0.345	5.61	8.76
E			0.285	0.325	7.24	8.25
F			.....	0.425	.....	10.79
G			0.850	0.880	21.59	22.35
H			38°	52°	38°	52°
J			0.260	0.290	6.60	7.36
K			0.148	0.154	3.76	3.91
M			.....	0.150	.....	3.55
N			.....	0.040	.....	1.01
P			1.500	1.750	38.10	44.45
R			0.325	.....	8.26	.....
S			.....	0.900	.....	22.86
T			.....	0.300	.....	7.62
V			0.200	.....	5.08	.....
Y			0.018	0.022	0.46	0.55
Z			0.038	0.042	0.97	1.06

NOTES

a. All dimensions in inches. Metric dimensions are furnished for information only.

b. This dimension shall be inspected periodically every 3 months during production. At start of production, a sample of six tubes, selected at random, shall meet acceptance requirements with no allowable defects. Failure of any dimension to meet acceptance requirements shall cause that dimension to be inspected with an AQL of 2.5 per cent individual and 4.0 per cent combined. When three successive samples as specified above meet acceptance requirements for this dimension, the dimension shall revert to a periodic inspection.

c. Antenna lead insulation shall be flush with the top of the outer coaxial conductor within ±.005 inch (0.13 mm) and be firmly fixed.

d. The outer extremity of the antenna lead shall be concentric with the center line of the outer coaxial conductor within ±.010 inch (0.25 mm).

e. Soldering to the heater leads in equipment is permissible if a minimum distance of ½ inch is maintained between the solder joint, or point of application of heat, and the glass button of the tube.

f. External ferrous parts, except the stainless steel parts, shall be plated with either silver, gold, nickel or other nonferrous material. No plating is required on the nonferrous external parts.

FIGURE 1. *Outline Drawing*—Continued

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